

Minimum observability of probabilistic Boolean networks ☆

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ABSTRACT

This paper studies the minimum observability of probabilistic Boolean networks (PBNs), the main objective of which is to add the fewest measurements such that an unobservable PBN becomes observable. First of all, the algebraic form of a PBN is established with the help of semi-tensor product (STP) of matrices. By combining the algebraic forms of two identical PBNs into a parallel system, a method to search the states that need to be H -distinguishable is proposed based on the robust set reachability technique. Secondly, a necessary and sufficient condition is given to find the minimum measurements such that a given set can be H -distinguishable. Moreover, by comparing the numbers of measurements for all the feasible H -distinguishable state sets, the least measurements that make the system observable are gained. Finally, an example is given to verify the validity of the obtained results.

1. Introduction

Boolean networks (BNs) are effective models to analyze gene regulatory networks in systems biology, which were first proposed by Kauffman in 1969 [1]. In a BN, each node can take “1” or “0” in a discrete time, indicating that the gene is open or closed. The evolution of each state variable is determined by the logical relationship among itself and its neighbor’s states. Since the BNs are the simplest logical dynamical systems, the research of BNs has aroused the attention of many experts and scholars. At present, BNs have been widely used in systems biology including cell differentiation [2], immune response [3], neural network [4]. In addition to biological networks, BNs have been successfully applied in many other fields, including networked evolutionary games [5], information security [6], wireless sensor networks [7], and so on.

In order to depict the randomness and uncertainty of gene regulatory networks, Shmulevich et al. proposed probabilistic Boolean networks (PBNs) [8]. The PBN is a random extension of deterministic BNs, and its state update follows a given probability distribution that controls the activation frequency of each subnetwork. PBN plays an important role both in theory and in the practical application of gene regulatory networks. A number of computational tools for the modeling and analysis of PBNs are developed, such as the mathematical theory of discrete-time Markov chains and integer programming approach. The dynamic behavior of PBNs can be modeled as Markov Chains. [9] dealt with the problem of optimal intervention for PBNs with the help of discrete-time Markov decision process. Integer programming can effectively solve the control problem of PBNs, such as the optimal external control [10].

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Among theoretical topics, [11] studied steady-state analysis of PBNs by determining the number of iterations required to make the Markov chain converge. [12] solved the reachability problem of PBNs by a model checking approach.

Due to the lack of effective research tools, it was very difficult to systematically analyze the dynamics of BNs and PBNs. The control problems of logical dynamic systems were studied even less. This situation was not solved until Cheng proposed the semi-tensor product (STP) of matrices [13]. Using the STP method, the logical forms of BNs and PBNs can be transformed into equivalent algebraic forms, which is convenient for us to analyze and manipulate these logical networks according to the classical control theory [14]. Up to now, many material problems of BNs and PBNs have been studied and solved via STP, such as controllability [15,16], observability [17,18], set stabilization and set stability [19–22], disturbance decoupling [23,24], and state estimation [25,26]. In addition to logical dynamic systems, STP has also been applied to many other fields, such as game theory [27,28], finite automata [29,30], function perturbations [31,32] and so on.

Observability is a fundamental and important concept in system science and control theory, which elaborates a problem of whether the initial state can be identified uniquely through the observed output sequences. In a BN, if any two different initial states can be distinguished by their output sequences, then the BN is observable. The existing results provide many methods to study the observability of BNs, such as the graph-theoretic approach [33], observability matrix-based analysis [34], set controllability method [35], finite automata approach [36] and the weighted-pair-graph method [37]. Since the BN is a deterministic system, the state trajectory of each given initial state is unique. For a PBN, due to the uncertainty of the evolutionary dynamics, the output sequence of any given initial state can not be completely determined. Thus, compared with deterministic BNs, the observability of PBNs is more challenging. Several kinds of observability about PBNs have been provided [38,39]. In [38], the definition of finite-time observability in probabilistic is given, which requires that for any two different initial states, the probability that their output sequences are not equal in finite time is nonzero. Two different concepts of observability are defined in [39]. One is finite-time observability with probability one, which requires that the probability of the corresponding output sequences being not equal in finite time is one for any two different initial states. The other case is asymptotic observability in distribution, which satisfies that for any two different initial states, when the time tends to infinity, the probability that the output sequences are different tends to one.

When a given system cannot be observed by its original measurements, it is necessary to add some new measurements to make it observable. Considering the costs, the number of additional sensors required is crucial. How to find the minimum number of measurements to make the system observable, called the minimum observability problem of the system, is a problem worth exploring. In logical dynamic systems, the minimum observability problem of conjunctive BNs is first studied. Reference [40] designs an effective algorithm to solve the minimum observability using graph theory approach. Subsequently, Liu. Y proposes a new method to study the minimum observability about general BNs, and obtains a method for determining the least states in which observations can be directly added [41]. Furthermore, [42] generalizes the result of minimum observability in [41] to the case with control inputs. Compared with deterministic BNs, the stochastic model is more in line with real life. However, as far as we know, there is no result about the minimum observability of PBNs.

In this paper, we investigate the minimum observability of PBNs for the first time. To solve the minimum observability of PBNs, we need to address two key issues: (i) determine which state sets must be H -distinguishable to make an unobservable PBN observable; (ii) add the least measurements to make a given state set H -distinguishable. For an unobservable PBN, the distinguishable states are not affected by the newly added observations. Therefore, this paper mainly studies the dynamic trajectories of indistinguishable states. In [41,42], the states that need to become initially distinguishable are determined by the attractors of the BNs and BCNs. However, since the state trajectories of PBNs are too complex, the method of [41,42] is not appropriate for PBNs. We study the minimum observability of PBNs mainly based on the robust reachability technique. The method in this paper is more general, and is also applicable to deterministic BNs. The main contributions of this paper are as follows.

(i) A method to make an unobservable PBN observable is proposed. A class of minimal state sets are obtained by the robust set reachability technique. As long as one of these sets becomes H -distinguishable, the PBN becomes observable.

(ii) The minimum observability problem of PBN is solved. By constructing a group of truth value matrices, all the schemes (including the optimal scheme) that make the given sets H -distinguishable are obtained. By comparing the numbers of measurements for the feasible H -distinguishable state sets, the least measurements added to make the system observable is determined.

The rest of this article is organized as follows. Part 2 contains some commonly used symbols and preliminaries. Main results of this article are in Part 3. An illustrative example is given in Part 4. Some brief conclusions are provided in Part 5.

2. Preliminaries

2.1. Symbols and STP of matrices

We first give some commonly used symbols in Table 1.

Definition 2.1. [13] The STP of $C \in \mathbb{R}_{a \times b}$ and $D \in \mathbb{R}_{q \times r}$ is defined as

$$C \ltimes D = (C \otimes I_{\frac{\lambda}{b}})(D \otimes I_{\frac{\lambda}{q}}),$$

where $\lambda = lcm(b, q)$.

Table 1
Notations.

Notations	Definitions
$\mathcal{D}$	$\{1, 0\}$
$\mathcal{D}^n$	$\underbrace{\mathcal{D} \times \cdots \times \mathcal{D}}_n$
$\mathbf{Z}_+$	set of positive integers
$\mathbb{R}_{m \times n}$	set of $m \times n$ real matrices
$\text{Col}_i(M)$	the i -th column of matrix M .
$\text{Row}_i(M)$	the i -th row of matrix M
δ_n^r	$\text{Col}_r(I_n)$
Δ_n	$\{\delta_n^r \mid r = 1, \dots, n\}$
$\text{Col}(M) \subseteq \Delta_m$	$M \in \mathbb{R}_{m \times n}$ is a logical matrix
$\mathcal{L}_{m \times n}$	set of $m \times n$ logical matrices
$L = \delta_m[i_1 \ i_2 \ \cdots \ i_n]$	matrix L with $\text{Col}_i(L) = \delta_m^{i_i}$
$L_{i,j}$	the (i, j) -element of matrix L
$ \cdot $	the cardinality of a set
$[L]_{i,j} \in \{0, 1\}$	L is a Boolean matrix.
$\mathcal{B}_{m \times n}$	set of $m \times n$ Boolean matrices
$M +_{\mathcal{B}} N$	$[M +_{\mathcal{B}} N]_{i,j} = [M]_{i,j} \vee [N]_{i,j}$
$L^\top$	the transpose of matrix L
$\mathbf{1}_n$	n -dimensional column vector whose entries are equal to 1.
$\text{lcm}(a_1, a_2, \dots, a_s)$	the least common multiple of $a_1, a_2, \dots, a_s$
$a \in \mathcal{A} \setminus \mathcal{B}$	$a \in \mathcal{A}, a \notin \mathcal{B}$
$\otimes$	Kronecker product of matrices
$*$	Khatri-Rao product of matrices

Remark 2.2. The STP is a generalization of the traditional matrix multiplication, which preserves almost all the properties of traditional matrix multiplication, thus “ $\bowtie$ ” can be omitted in the following.

Definition 2.3. [14] The Khatri-Rao product of $M \in \mathcal{M}_{m \times r}$ and $N \in \mathcal{M}_{n \times r}$ is defined as

$$M * N = [\text{Col}_1(M) \otimes \text{Col}_1(N), \text{Col}_2(M) \otimes \text{Col}_2(N), \dots, \text{Col}_r(M) \otimes \text{Col}_r(N)].$$

Identify “ $1 \sim \delta_2^1$ ” and “ $0 \sim \delta_2^2$ ”, where “ $\sim$ ” represents an equivalence relation, then $\Delta \sim \mathcal{D}$. We call δ_2^1 and δ_2^2 the vector form of 1 and 0, respectively. Based on the vector form of the logical variables, a Boolean function can be expressed as equivalent algebraic form.

Lemma 2.4. [13] Let $f : \mathcal{D}^r \rightarrow \mathcal{D}$ be a Boolean mapping, then there exists a unique $M_f \in \mathcal{L}_{2 \times 2^r}$ such that

$$f(X_1, X_2, \dots, X_r) = M_f \bowtie_{i=1}^r x_i,$$

where $x_i \in \Delta$, $i = 1, 2, \dots, r$ is the vector form of $X_i \in \mathcal{D}$, and M_f is the structure matrix of f .

2.2. Algebraic representation of PBNs

A PBN with n state nodes and q output nodes is described as

$$\begin{cases} x_i(t+1) = f_i^{\sigma(t)}(x_1(t), x_2(t), \dots, x_n(t)), & i = 1, 2, \dots, n, \\ y_j(t) = h_j(x_1(t), x_2(t), \dots, x_n(t)), & j = 1, 2, \dots, q, \end{cases} \quad (2.1)$$

where $x_i(t) \in \mathcal{D}$ and $y_j(t) \in \mathcal{D}$ are the state and the output of the system (2.1) at time t , respectively. f_i^σ and h_j are logical functions. The switching signal $\sigma(t)$ is a stochastic sequence taking values from $[1 : m]$, where m is a positive integer denoting the number of subnetworks.

Using the vector form of variables and Lemma (2.4), PBN (2.1) can be represented by

$$\begin{cases} x(t+1) = \mathbf{L}_{\sigma(t)} x(t), \\ y(t) = H x(t), \end{cases} \quad (2.2)$$

where $x(t) = \bowtie_{i=1}^n x_i(t) \in \Delta_{2^n}$, $y(t) \in \Delta_{2^q}$, and $\mathbf{L}_v \in \mathcal{L}_{2^n \times 2^n}$, $v \in [1 : m]$ is the structure matrix of the v -th sub-network. Identify $v \sim \delta_m^v$, where $v \in [1 : m]$, then PBN (2.2) can be expressed equivalently as

$$\begin{cases} x(t+1) = \mathbf{L} \sigma(t) x(t), \\ y(t) = H x(t), \end{cases} \quad (2.3)$$

where $\sigma(t) \in \Delta_m$ and $\mathbf{L} = [\mathbf{L}_1 \ \mathbf{L}_2 \ \dots \ \mathbf{L}_m]$.

Suppose that the stochastic switching signal $\sigma(t)$ is an independent and identically distributed process with a probability distribution of $\mathbb{P}\{\sigma(t) = \delta_m^i\} = p_i^\sigma, i \in [1 : m]$, where $0 \leq p_i^\sigma \leq 1$ and $\sum_{i=1}^m p_i^\sigma = 1$. Denote the probability distribution vector of $\sigma(t)$ as $\mathbf{p}^\sigma = [p_1^\sigma \ p_2^\sigma \ \dots \ p_m^\sigma]^\top$, then the one-step state transition probability matrix of PBN (2.3) can be expressed as

$$\mathbf{P}^x = \sum_{i=1}^m p_i \mathbf{L}_i = \mathbf{L} \ltimes \mathbf{p}^\sigma, \quad (2.4)$$

where $\mathbf{P}_{i,j}^x = \mathbb{P}\{x(t+1) = \delta_{2n}^j | x(t) = \delta_{2n}^i\}$.

Let σ^s represent a set of switching signal sequence with length s , $\sigma^s = (\sigma(0), \sigma(1), \dots, \sigma(s-1))$, and denote the set of all switching signal sequences with length s as Σ^s . The state of system (2.2) at time s starting from x_0 under switching signal $\sigma^s \in \Sigma^s$ is represented by $x(s; \sigma^s, x_0)$ and the corresponding output is represented by $y(s; \sigma^s, x_0)$, i.e., $y(s; \sigma^s, x_0) = Hx(s; \sigma^s, x_0)$.

2.3. Observability of PBNs

Definition 2.5. [39] PBN (2.3) is observable with probability one in finite time, if for any distinct initial state pair $x_0, \bar{x}_0 \in \Delta_{2n}$, there exists a positive integer T such that

$$\mathbb{P}\{(y_0, y(1; \sigma^1, x_0), \dots, y(T; \sigma^T, x_0)) \neq (y_0, y(1; \sigma^1, \bar{x}_0), \dots, y(T; \sigma^T, \bar{x}_0))\} = 1. \quad (2.5)$$

These two states $x_0, \bar{x}_0$ are called distinguishable states. Otherwise, they are indistinguishable states.

With the parallel extension method, the observability problem of PBN (2.3) can be transformed into the set reachability problem of an augmented system. The main steps can be summarized as follows.

First of all, a duplicate of PBN (2.3) is copied as

$$\begin{cases} x'(t+1) = \mathbf{L}\sigma(t)x'(t), \\ y'(t) = Hx'(t). \end{cases} \quad (2.6)$$

Let $z(t) := x(t) \ltimes x'(t)$, $w(t) := y(t) \ltimes y'(t)$, an augmented system can be constructed as

$$\begin{aligned} z(t+1) &= x(t+1) \ltimes x'(t+1) \\ &= (\mathbf{L}\sigma(t)x(t)) \ltimes (\mathbf{L}\sigma(t)x'(t)) \\ &= [\mathbf{L}(I_{m2n} \otimes \mathbf{1}_{2n}^\top) \sigma(t)x(t)x'(t)] \ltimes [\mathbf{L}(I_m \otimes \mathbf{1}_{2n}^\top) \sigma(t)x(t)x'(t)] \\ &= \{[\mathbf{L}(I_{m2n} \otimes \mathbf{1}_{2n}^\top)] * [\mathbf{L}(I_m \otimes \mathbf{1}_{2n}^\top)]\} \sigma(t)z(t) \\ &:= \mathbf{F}\sigma(t)z(t), \\ w(t) &= Hx(t)Hx'(t) \\ &= H(I_{2n} \otimes H)x(t)x'(t) \\ &= (H \otimes I_{2n})(I_{2n} \otimes H)z(t) \\ &= (H \otimes H)z(t). \end{aligned} \quad (2.7)$$

Divide $\mathbf{F}$ into m equal blocks according to the dimension of switching signal as $\mathbf{F} = [\mathbf{F}_1 \ \mathbf{F}_2 \ \dots \ \mathbf{F}_m]$, where $\mathbf{F}_i \in \mathcal{L}_{2^{2n} \times 2^{2n}}, i \in [1 : m]$. Let $\mathbf{Q} = \sum_{i=1}^m p_i^\sigma \mathbf{F}_i = \mathbf{F} \ltimes \mathbf{p}^\sigma$ and $\mathbf{K} = H \otimes H \in \mathcal{L}_{2^{2q} \times 2^{2n}}$, then the expectation of the augmented system (2.7) can be expressed as

$$\begin{cases} \mathbb{E}\{z(t+1)\} = \mathbf{Q}\mathbb{E}\{z(t)\}, \\ w(t) = \mathbf{K}z(t). \end{cases} \quad (2.8)$$

Obviously, for any $z_0 = x_0 \ltimes x'_0$, we have

$$z(t; \sigma^t, z_0) = x(t; \sigma^t, x_0) \ltimes x'(t; \sigma^t, x'_0),$$

where $z(t; \sigma^t, z_0)$ represents the state starting from initial state z_0 of system (2.8) at time $t \in \mathbb{Z}_+$ under switching sequence $\sigma^t \in \Sigma^t$.

Definition 2.6. Consider two states $x, x' \in \Delta_{2n}$ and $z = x \ltimes x' \in \Delta_{2^{2n}}$. z is said to be distinguishable, if x, x' are distinguishable.

For two state sets $\mathcal{Z}_1 = \{\delta_{2n}^i \delta_{2n}^j \mid i < j\}$ and $\mathcal{Z}_2 = \{\delta_{2n}^j \delta_{2n}^i \mid i < j\}$. If $\mathcal{Z}_1$ is a distinguishable set, then $\mathcal{Z}_2$ is distinguishable, otherwise, $\mathcal{Z}_2$ is indistinguishable, i.e., $\mathcal{Z}_1$ and $\mathcal{Z}_2$ have the same distinguishability. Therefore, in order to decrease computational complexity, we only detect the distinguishability of $\mathcal{Z}_1$. Then three subsets are defined as

$$\begin{cases} S_0 = \{\delta_{2^n}^i \delta_{2^n}^j \mid i = 1, 2, \dots, 2^n\}, \\ S_1 = \{\delta_{2^n}^i \delta_{2^n}^j \mid H \delta_{2^n}^i = H \delta_{2^n}^j, i < j\}, \\ S_2 = \{\delta_{2^n}^i \delta_{2^n}^j \mid H \delta_{2^n}^i \neq H \delta_{2^n}^j, i < j\}, \end{cases} \quad (2.9)$$

where S_0 is the diagonal subset, S_1 is the H -indistinguishable subset, and S_2 is the H -distinguishable subset.

Subsequently, the observability of PBN (2.3) is explored based on the set reachability of PBNs [39].

Definition 2.7. [39] Suppose that $C_0, C_d \subset \Delta_{2^{2n}}$ are the initial and target sets, respectively. C_d is said to be finite-time reachable with probability one from C_0 , if there exists a positive integer k such that for any initial state $z_0 \in C_0$, it holds that

$$\mathbb{P}\{z(k; \sigma^k, z_0) \in C_d\} = 1$$

for any $\sigma^k \in \Sigma^k$.

Lemma 2.8. [39] PBN (2.3) is observable with probability one in finite time, if and only if the set S_2 is reachable from S_1 with probability one in finite time in system (2.8).

Lemma 2.8 converts the observability of PBN (2.3) into the set reachability problem of augmented system (2.8). If there exists one state $\delta_{2^{2n}}^s \in S_1$ such that $\delta_{2^{2n}}^s$ can not reach S_2 , then PBN (2.3) is unobservable, and $\delta_{2^{2n}}^s$ is an indistinguishable state. In the next section, we consider to add the least measurements to make PBN (2.3) observable.

3. Main results

3.1. The method to make a PBN observable

In this subsection, we consider to solve the first issue, that is, search the sets that need to be H -distinguishable such that an unobservable PBN can become observable.

For PBN (2.3), suppose that for any $j_a \in \{1, 2, \dots, n\}$, x_{j_a} can be directly measurable under the new measurement $y_{q+a} = x_{j_a}$. We study the observability of PBNs via STP. That is to say, a new measurement set $I = \{x_{j_1}, x_{j_2}, \dots, x_{j_m}\}$ is added to make the original unobservable system observable. Denote the added measurements as

$$y_{q+1}(t) = x_{j_1}(t), \dots, y_{q+m}(t) = x_{j_m}(t). \quad (3.1)$$

Let $\bar{y}(t) = y_{q+1} \bowtie y_{q+2} \bowtie \dots \bowtie y_{q+m}$, the added outputs can be represented by $\bar{y}(t) = \bar{H}x(t)$. Further, the new outputs can be depicted as

$$\begin{cases} y_j(t) = h_j(x_1(t), x_2(t), \dots, x_n(t)), j = 1, \dots, q, \\ y_{q+1}(t) = x_{j_1}(t), \\ y_{q+2}(t) = x_{j_2}(t), \\ \vdots \\ y_{q+m}(t) = x_{j_m}(t). \end{cases} \quad (3.2)$$

Denote $y^*(t) = y_1 \bowtie y_2 \bowtie \dots \bowtie y_{q+m}$, system (3.2) can be represented by $y^*(t) = Gx(t)$, where $G = H * \bar{H}$.

Evidently, if $x, x' \in \Delta_{2^n}$ are distinguishable under original output, then x, x' are distinguishable under the new output. Thus, we only need to make the indistinguishable states become distinguishable under new added measurements.

Lemma 3.1. [42] Suppose that $z = x \bowtie x'$ is indistinguishable in original PBN (2.3), and $x = \bowtie_{r=1}^n \delta_{2^n}^{i_r}$, $x' = \bowtie_{r=1}^n \delta_{2^n}^{j_r}$. If $\delta_2^{i_k} \neq \delta_2^{j_k}$, then $x, x' (z)$ will become H -distinguishable under new added measurement $y(t) = x_k(t)$.

According to Lemma 2.8, for an unobservable PBN, there exists at least one indistinguishable state in S_1 which can not reach S_2 . We need to find all these states and make them distinguishable. We first calculate the states in S_1 that can reach S_2 with probability one. We first give the following definition.

Definition 3.2. For a given set $\Lambda \subseteq \Delta_{2^{2n}}$, define its index vector $\mathbf{J}_\Lambda \in B_{2^{2n} \times 1}$ as follows:

$$(\mathbf{J}_\Lambda)_i = \begin{cases} 1, & \delta_{2^{2n}}^i \in \Lambda, \\ 0, & \delta_{2^{2n}}^i \notin \Lambda. \end{cases}$$

In the following, we calculate the reachable set at each step of S_2 based on the vector representation of sets. Calculate a set

$$R_1(S_2) = \{\delta_{2^{2n}}^j \mid \mathbf{J}_{S_2}^\top \mathbf{Q} \delta_{2^{2n}}^j = 1, \delta_{2^{2n}}^j \notin S_2\}, \quad (3.3)$$

where $\mathbf{J}_{S_2}$ is the index vector of set S_2 . For any $\delta_{2^{2n}}^j \in R_1(S_2)$, it holds that

$$\begin{aligned} 1 &= \mathbf{J}_{S_2}^\top \mathbf{Q} \delta_{2^{2n}}^j \\ &= \sum_{\delta_{2^{2n}}^i \in S_2} \mathbf{Q}_{i,j} \\ &= \sum_{\delta_{2^{2n}}^i \in S_2} \mathbb{P}\{z(t+1) = \delta_{2^{2n}}^i | z(t) = \delta_{2^{2n}}^j\} \\ &= \mathbb{P}\{z(t+1) \in S_2 | z(t) = \delta_{2^{2n}}^j\}. \end{aligned}$$

Thus $\delta_{2^{2n}}^j$ can reach S_2 with probability one in one step. It is obvious that $R_1(S_2)$ is the one-step reachable set of S_2 with probability one.

Similarly, we can calculate set

$$\begin{aligned} R_2(S_2) &= R_1(S_2 \cup R_1(S_2)) \\ &= \{\delta_{2^{2n}}^j | (\mathbf{J}_{S_2}^\top + \mathbf{J}_{R_1(S_2)}^\top) \mathbf{Q} \delta_{2^{2n}}^j = 1, \delta_{2^{2n}}^j \notin S_2 \cup R_1(S_2)\}. \end{aligned}$$

For any $\delta_{2^{2n}}^j \in R_2(S_2)$, one has

$$\begin{aligned} 1 &= (\mathbf{J}_{S_2}^\top + \mathbf{J}_{R_1(S_2)}^\top) \mathbf{Q} \delta_{2^{2n}}^j \\ &= \sum_{\delta_{2^{2n}}^i \in S_2 \cup R_1(S_2)} \mathbf{Q}_{i,j} \\ &= \sum_{\delta_{2^{2n}}^i \in S_2 \cup R_1(S_2)} \mathbb{P}\{z(t+1) = \delta_{2^{2n}}^i | z(t) = \delta_{2^{2n}}^j\} \\ &= \mathbb{P}\{z(t+1) \in S_2 \cup R_1(S_2) | z(t) = \delta_{2^{2n}}^j\} \\ &= \mathbb{P}\{z(t+2) \in S_2 | z(t) = \delta_{2^{2n}}^j\}. \end{aligned}$$

We can easily obtain that $R_2(S_2)$ is the two-step reachable set of S_2 with probability one. Since $\delta_{2^{2n}}^j \notin S_2 \cup R_1(S_2)$, $\delta_{2^{2n}}^j$ can not reach S_2 in one step with probability one.

Recursively, we can obtain a group of sets

$$\begin{aligned} R_k(S_2) &= R_1(S_2 \cup R_1(S_2) \cup \dots \cup R_{k-1}(S_2)) \\ &= \{\delta_{2^{2n}}^j | (\sum_{i=1}^{k-1} \mathbf{J}_{R_i(S_2)}^\top + \mathbf{J}_{S_2}^\top) \mathbf{Q} \delta_{2^{2n}}^j = 1, \delta_{2^{2n}}^j \notin S_2 \cup R_1(S_2) \cup \dots \cup R_{k-1}(S_2)\}, \text{ where } k > 2. \end{aligned} \quad (3.4)$$

Obviously, $R_k(S_2)$ is the k -step reachable set of S_2 with probability one.

Since the number of states of a PBN is finite, there must exist a positive integer $t^* \leq 2^{2n}$ such that $R_{t^*}(S_2) \neq \emptyset$ and $R_{t^*+1}(S_2) = \emptyset$. We can get Algorithm 1.

Algorithm 1 Calculation of the reachable set of S_2 with probability one.

Require: $\mathbf{J}_{S_2}, \mathbf{Q}$

Ensure: $\bigcup_{i=1}^{t^*} R_i(S_2)$

```

1: Initialize:  $k$ ;
2: Calculate  $R_1(S_2)$  by (3.3);
3: if  $R_1(S_2) = \emptyset$  stop
4: else Go to 5;
5: while  $k \in [2 : 2^{2n}]$  do
6: Calculate  $R_k(S_2)$  by (3.4);
7: if  $R_{k-1}(S_2) \neq \emptyset, R_k(S_2) = \emptyset$  then
8:  $R_{t^*}(S_2) = R_k(S_2)$ ;
9: else  $k = k + 1$ ;
10: Go to 5;
11: end if
12: end while
```

Remark 3.3. Algorithm 1 shows that the set of all states that can reach S_2 with probability one is $\bigcup_{i=1}^{t^*} R_i(S_2)$. Denote $\mathcal{M} := [\bigcup_{i=1}^{t^*} R_i(S_2)] \cap S_1$, then $\mathcal{M}$ contains all the states in S_1 which are distinguishable.

Let $\mathcal{N} = S_1 \setminus \mathcal{M}$, then $\mathcal{N}$ denotes the set of indistinguishable states in S_1 . To make the system become observable, new measurements must be added to make the states in $\mathcal{N}$ distinguishable. Adding measurements directly to make all states in $\mathcal{N}$ be

H -distinguishable is one method, but it may add a lot of redundant measurements. Since a measurement can make multiple states H -distinguishable, the measurements that need to be added can not be determined just by the number of states. In the following, we search all the feasible minimal sets that need to be H -distinguishable. The system becomes observable only if all states in a minimal set are H -distinguishable.

Define a state set in $\mathcal{N}$ as follows:

$$\mathcal{N}_1 = \{\delta_{2^{2n}}^j | \mathbf{J}_{S_0}^\top \mathbf{Q} \delta_{2^{2n}}^j > 0, \delta_{2^{2n}}^j \in \mathcal{N}\}. \quad (3.5)$$

$\mathbf{J}_{S_0}^\top \mathbf{Q} \delta_{2^{2n}}^j > 0$ shows that there exists at least one switching signal such that $\delta_{2^{2n}}^j$ can reach S_0 . Thus, $\mathcal{N}_1$ contains all the states in $\mathcal{N}$ that can reach S_0 in one step under at least one switching signal.

For set $\mathcal{N}$, we can find some states with the following characteristics: each state can reach itself in one step under at least one switching signal. The definition of such a state is given below.

Definition 3.4. [43] A state $z_0 \in \Delta_{2^{2n}}$ is called a positive-probability fixed point of PBN (2.8), if $\mathbb{P}\{z(t+1) = z_0 | z(t) = z_0\} > 0, \forall t \in \mathbb{N}$.

For state $\delta_{2^{2n}}^j \in \mathcal{N}$, if

$$\mathbf{Q}_{jj} > 0, \quad (3.6)$$

then $\delta_{2^{2n}}^j$ is a positive-probability fixed point in set $\mathcal{N}$. Based on formula (3.6), we can easily calculate all the positive-probability fixed points in set $\mathcal{N}$ and denoted it as

$$\mathcal{P} = \{\delta_{2^{2n}}^j \in \mathcal{N} | \mathbf{Q}_{jj} > 0\}. \quad (3.7)$$

Lemma 3.5. To make PBN (2.3) be observable, $\mathcal{N}_1 \cup \mathcal{P}$ must be H -distinguishable.

The proof of Lemma 3.5 and all other proofs can be found in Appendix.

Let $\Gamma = \mathcal{N}_1 \cup \mathcal{P} \cup S_2$. To make PBN (2.3) observable, Γ should become H -distinguishable by adding new measurements. According to Algorithm 1, we can calculate the reachable set of Γ with probability one and denote it as $R(\Gamma)$. Obviously, $\mathcal{M} \subseteq R(\Gamma)$. Since S_2 is an H -distinguishable set, if $\mathcal{N}_1 \cup \mathcal{P}$ is also H -distinguishable, then $R(\Gamma)$ must be distinguishable.

Furtherly, if $\mathcal{N} \setminus (\mathcal{N}_1 \cup \mathcal{P}) \subseteq R(\Gamma)$, then PBN (2.1) becomes observable. Otherwise, let

$$\mathcal{N}' = \mathcal{N} \setminus [\mathcal{N}_1 \cup \mathcal{P} \cup R(\Gamma)], \quad (3.8)$$

then $\mathcal{N}'$ is still an indistinguishable set. We need to add appropriate measurements to make $\mathcal{N}'$ distinguishable.

However, $\mathcal{N}'$ just needs to be a distinguishable set. To make $\mathcal{N}'$ distinguishable, we do not need to make all the states in it H -distinguishable. We hope to find a subset $\Theta \subseteq \mathcal{N}'$ such that

$$\mathcal{N}' \setminus \Theta \subseteq R(\Theta \cup \Gamma), \quad (3.9)$$

where $R(\Theta \cup \Gamma)$ denotes the reachable set of $\Theta \cup \Gamma$ with probability one. That is, we need to find a subset Θ from $\mathcal{N}'$ such that the other states of $\mathcal{N}'$ will reach $\Theta \cup \Gamma$ with probability one. If Θ is distinguishable, then $\mathcal{N}'$ must be distinguishable. So we just need to make such a set Θ H -distinguishable.

It is a difficult task to find all the sets Θ that satisfy formula (3.9). We can directly find all the nonempty subsets of $\mathcal{N}'$, and then detect which subset satisfies formula (3.9) one by one. Suppose that $\mathcal{N}'$ has γ states and denote all its nonempty subsets by $\Theta_1, \Theta_2, \dots, \Theta_{2^\gamma-1}$. According to Algorithm 1 and formula (3.9), we can find all sets $\Theta_i, i \in [1 : 2^\gamma - 1]$ which make $\mathcal{N}' \setminus \Theta_i$ reach $\Theta_i \cup \Gamma$ with probability one. However, the computational complexity of this method is very high, we need to determine whether $2^\gamma - 1$ sets satisfy formula (3.9). To reduce the computational complexity, it is necessary to search the states that must become H -distinguishable from $\mathcal{N}'$.

If $\mathcal{N}'$ has an invariant set, to make $\mathcal{N}'$ distinguishable, its invariant set must be distinguishable.

Definition 3.6. [22] A nonempty set $C \subseteq \Delta_{2^{2n}}$ is called an invariant set with probability one of PBN (2.8), if for any $z(t) \in C$, it holds that

$$\mathbb{P}\{z(t+1) \in C | z(t) \in C\} = 1. \quad (3.10)$$

The union of all invariant sets contained in a given set $C \subseteq \Delta_{2^{2n}}$ is called the maximum invariant set of C , denoted by $I(C)$.

There have been many excellent results on the calculation of the maximum invariant set of PBNs with probability one [22]. Referring to these results, we can calculate the maximum invariant set $I(\mathcal{N}')$ of $\mathcal{N}'$.

If we want to make $I(\mathcal{N}')$ distinguishable, there is no need to make all the states of $I(\mathcal{N}')$ H -distinguishable. We can first calculate the state set that needs to become H -distinguishable in $I(\mathcal{N}')$.

Since $I(\mathcal{N}')$ is an invariant set, there must be a subset in $I(\mathcal{N}')$ such that all the other states in $I(\mathcal{N}')$ will reach the subset with probability one. Suppose that there are α states in $I(\mathcal{N}')$, then $I(\mathcal{N}')$ has $2^\alpha - 1$ nonempty subsets and denote them as $\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_{2^\alpha-1}$. According to Algorithm 1, the reachable set of each set $\mathcal{G}_i, i = [1 : 2^\alpha - 1]$ with probability one can be calculated respectively. Then the state set that can reach $\mathcal{G}_i$ with probability one is $R(\mathcal{G}_i)$. If

$$I(\mathcal{N}') \setminus \mathcal{G}_i \subseteq R(\mathcal{G}_i), \quad (3.11)$$

then $\mathcal{G}_i$ is a subset to make $I(\mathcal{N}') \setminus \mathcal{G}_i$ reach $\mathcal{G}_i$ with probability one. We denote the complete set of all sets satisfying (3.7) as Ω .

Algorithm 2 is given to find set Ω , where Ω contains all sets $\mathcal{G}_i$ in $I(\mathcal{N}')$ which make $I(\mathcal{N}') \setminus \mathcal{G}_i$ reach $\mathcal{G}_i$ with probability one.

Algorithm 2 The solution method of Ω in $I(\mathcal{N}')$.

Require: $I(\mathcal{N}')$

Ensure: Ω

```

1: Calculate sets  $\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_{2^\alpha-1}$ ;
2: Calculate set  $R(\mathcal{G}_i), i = [1 : 2^\alpha - 1]$ ;
3: Initialize:  $\Omega = \emptyset$ ;
4: for  $i = [1 : 2^\alpha - 1]$  do
5:   if  $I(\mathcal{N}') \setminus \mathcal{G}_i \subseteq R(\mathcal{G}_i)$  then
6:      $\Omega = \{\Omega, \mathcal{G}_i\}$ ;
7:   end if
8: end for
```

We continue to observe the relationship among the sets in Ω . For any sets $\mathcal{G}_i, \mathcal{G}_j \in \Omega$, if $\mathcal{G}_i \subseteq \mathcal{G}_j$, we will delete $\mathcal{G}_i$ from set Ω . After the above operation, assume that Ω remains s sets that are not included in each other, denoted by $\Omega_1, \Omega_2, \dots, \Omega_s$. Then all states in $I(\mathcal{N}') \setminus \Omega_v$ can reach $\Omega_v, v \in [1 : s]$ with probability one. If any a set of $\Omega_1, \Omega_2, \dots, \Omega_s$ is H -distinguishable, then $I(\mathcal{N}')$ must be distinguishable.

When $I(\mathcal{N}')$ is distinguishable, we determine whether formula $\mathcal{N}' \setminus I(\mathcal{N}') \subseteq R(\Gamma \cup I(\mathcal{N}'))$ is valid. If the formula holds, then PBN (2.1) is observable. Otherwise, let

$$\mathcal{N}'' = \mathcal{N}' \setminus R(\Gamma \cup I(\mathcal{N}')). \quad (3.12)$$

$\mathcal{N}''$ contains the states that are still indistinguishable in $\mathcal{N}'$ even after $\mathcal{N}'_1 \cup \mathcal{P}$ and $I(\mathcal{N}')$ all become distinguishable. We continue to search the states that need to become H -distinguishable so that $\mathcal{N}''$ becomes distinguishable.

Suppose that there are β states in $\mathcal{N}''$, then $\mathcal{N}''$ has $2^\beta - 1$ nonempty subsets and denote them as $\bar{\mathcal{G}}_1, \bar{\mathcal{G}}_2, \dots, \bar{\mathcal{G}}_{2^\beta-1}$. Based on Algorithm 1, we can calculate a set Ω' , where Ω' contains all sets $\bar{\mathcal{G}}_i$ in $\mathcal{N}''$ which make $\mathcal{N}'' \setminus \bar{\mathcal{G}}_i$ reach $\Gamma \cup R(\Gamma \cup I(\mathcal{N}')) \cup \bar{\mathcal{G}}_i$ with probability one, $i \in [0 : 2^\beta - 1]$. For any sets $\bar{\mathcal{G}}_i, \bar{\mathcal{G}}_j \in \Omega'$, if $\bar{\mathcal{G}}_i \subseteq \bar{\mathcal{G}}_j$, we will delete $\bar{\mathcal{G}}_i$ from set Ω' . Assume that Ω' remains c sets that are not included in each other and denote them by $\Omega'_1, \Omega'_2, \dots, \Omega'_c$. If any a set of $\Omega'_1, \Omega'_2, \dots, \Omega'_c$ is H -distinguishable, then $\mathcal{N}''$ must be distinguishable. Define

$$\Theta_v^w = \Omega_v \cup \Omega'_w, \quad (3.13)$$

where $v \in [1 : s], w \in [1 : c]$. It is easy to see that set Θ_v^w satisfies condition (3.9), that is, all states in $\mathcal{N}' \setminus \Theta_v^w$ can reach $\Theta_v^w \cup \Gamma$ with probability one. If set Θ_v^w is H -distinguishable, then $\mathcal{N}'$ must be distinguishable, $v \in [1 : s], w \in [1 : c]$. Therefore, Θ_v^w is the set that we want.

Theorem 3.7. PBN (2.3) becomes observable, if and only if at least one set $\mathcal{N}'_1 \cup \mathcal{P} \cup \Theta_v^w$ becomes H -distinguishable, $v \in [1 : s], w \in [1 : c]$.

Based on the above analysis, we have the following key steps to determine all the feasible minimal state sets that need to be H -distinguishable.

- Step 1: Calculate the state set $\mathcal{M}$ in set S_1 that can reach S_2 with probability one, and obtain the indistinguishable state set $\mathcal{N} = S_1 \setminus \mathcal{M}$.
- Step 2: Identify the state set $\mathcal{N}'_1 \cup \mathcal{P}$ in $\mathcal{N}$ which clearly needs to be H -distinguishable, where $\mathcal{N}'_1$ is the set that can reach S_0 in one step under at least one switching signal and $\mathcal{P}$ is the positive-probability fixed point set. Determine the state set $\mathcal{N}'$ which remains indistinguishable after $\mathcal{N}'_1 \cup \mathcal{P}$ becomes H -distinguishable. If $\mathcal{N}' = \emptyset$, PBN (2.3) becomes observable; otherwise, go to Step 3.
- Step 3: Calculate the maximum invariant set $I(\mathcal{N}')$ of $\mathcal{N}'$, and search the minimal state sets $\Omega_v, v \in [1 : s]$ in $I(\mathcal{N}')$ that need to be H -distinguishable. Obtain the state set $\mathcal{N}''$ which remains indistinguishable after $\mathcal{N}'_1 \cup \mathcal{P} \cup \Omega_v$ becomes H -distinguishable. If $\mathcal{N}'' = \emptyset$, PBN (2.3) becomes observable; otherwise, go to Step 4.
- Step 4: Determine the minimal state subsets $\Omega'_w, w \in [1 : c]$ in $\mathcal{N}''$ that need to be H -distinguishable until $\mathcal{N}'' \setminus \Omega'_w$ can reach the new H -distinguishable set with probability one. All the minimal state sets which need to be H -distinguishable are $\mathcal{N}'_1 \cup \mathcal{P} \cup \Omega_v \cup \Omega'_w, v \in [1 : s], w \in [1 : c]$.

Remark 3.8. According to the above process to search set Θ_v^w , we just need to detect whether $2^\alpha + 2^\beta - 2$ sets satisfy formula (3.9), where $\alpha + \beta \leq \gamma$. The computational complexity of this method is much lower than that of directly detecting all the $2^\gamma - 1$ nonempty subsets in $\mathcal{N}'$.

Remark 3.9. In previous studies, the states that need to be H -distinguishable are determined by the attractors of BNs. However, since the state trajectories of a PBN from an initial state are not unique, the previous methods can not be applied to our models any more. It is much more difficult to determine the minimal sets that need to be H -distinguishable of a PBN than that of a BN.

3.2. The minimum observability of PBNs

In this subsection, we consider to add the least measurements to make an unobservable PBN become observable.

Firstly, we discuss how to add the minimal number of measurements to make all states in a set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ H -distinguishable, $v \in [1 : s], w \in [1 : c]$.

We first discuss which states can become H -distinguishable when the new measurement $y(t) = x_m(t)$ is added. Using STP, $y(t) = x_m(t)$ can be represented by algebraic form $y(t) = H_m x(t)$, where $H_m = (\mathbf{1}_{2^{m-1}}^\top \otimes I_2)(I_{2^m} \otimes \mathbf{1}_{2^{n-m}}^\top)$. For state $z = \delta_{2^n}^i \delta_{2^n}^j$, if

$$H_m \delta_{2^n}^i \neq H_m \delta_{2^n}^j, \quad (3.14)$$

then state z becomes H -distinguishable under the new measurement $y(t) = x_m(t)$. According to formula (3.14), we can calculate a set Y_m whose states are H -distinguishable under the new measurement $y(t) = x_m(t)$, where $Y_m = \{\delta_{2^n}^i \delta_{2^n}^j \mid H_m \delta_{2^n}^i \neq H_m \delta_{2^n}^j\}$.

Based on the above calculation process, we can find the state sets becoming H -distinguishable under the new outputs $y = x_i, i \in [1 : n]$ respectively, and denoted them as $Y_1, Y_2, \dots, Y_n$.

Choose a set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ arbitrarily, $v \in [1 : s], w \in [1 : c]$. For state $z \in \mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$, if $z \in Y_i, i \in [1 : n]$, then z is H -distinguishable under new added output $y = x_i$. For any given state, we can find all the measurements that make it H -distinguishable. Assume that $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w = \{z_1, z_2, \dots, z_l\}$, we can define a Boolean vector $\xi_j \in \mathcal{B}_{n \times 1}$ as follows:

$$(\xi_j)_i = \begin{cases} 1, & \text{if } z_j \in Y_i, \\ 0, & \text{if } z_j \notin Y_i. \end{cases} \quad (3.15)$$

Construct a truth matrix $\Phi_v^w = [\xi_1, \xi_2, \dots, \xi_l], v \in [1 : s], w \in [1 : c]$. We can easily see that $(\Phi_v^w)_{i,j} = 1$ if and only if $z_j \in \mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ is distinguishable under output $y = x_i$.

Theorem 3.10. $\{x_{j_1}, x_{j_2}, \dots, x_{j_r}\}$ is the set with the minimum measurements that makes all states in $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ H -distinguishable, if and only if $\mathcal{I}_v^w = \{j_1, j_2, \dots, j_r\}$ is the set with the least number of elements in $\{1, 2, \dots, n\}$ such that

$$\bigvee_{i=1}^r \text{Row}_{j_i}(\Phi_v^w) = \mathbf{1}_l^\top, \quad (3.16)$$

where l is the number of elements of set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$.

Remark 3.11. Theorem 3.10 provides a method to search the least measurements that make a given set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w, v \in [1 : s], w \in [1 : c]$ H -distinguishable. However, the selection of Θ_v^w is random, so these measurements are not necessarily the optimal solution of the minimum observation problem for PBN (2.1).

Next we will discuss how to add the least amount of measurements to make PBN (2.1) observable. For a set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w, v \in [1 : s], w \in [1 : c]$, we can calculate all measurement sets which has the least number of elements by Theorem 3.10 and denote them as $(\mathcal{I}_v^w)_1, (\mathcal{I}_v^w)_2, \dots, (\mathcal{I}_v^w)_\beta$. Obviously,

$$|(\mathcal{I}_v^w)_1| = |(\mathcal{I}_v^w)_2| = \dots = |(\mathcal{I}_v^w)_\beta|.$$

Denote the cardinality of these sets as λ_v^w .

For all $v \in [1 : s], w \in [1 : c]$, we can compute the cardinalities of the minimum number of measurements that make set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ H -distinguishable, and denoted them as

$$\lambda_1^1, \lambda_1^2, \dots, \lambda_1^c, \lambda_2^1, \lambda_2^2, \dots, \lambda_2^c, \dots, \lambda_s^1, \lambda_s^2, \dots, \lambda_s^c.$$

If

$$(v^*, w^*) = \argmin_{v \in [1 : s], w \in [1 : c]} \lambda_v^w, \quad (3.17)$$

then as long as we make set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_{v^*}^{w^*}$ H -distinguishable, it can be guaranteed that the added measurements are the minimum. Suppose that $\mathcal{I}_{v^*}^{w^*} = \{j_1^*, j_2^*, \dots, j_r^*\}$ is the set with the least number of elements in $\{1, 2, \dots, n\}$ such that all states in $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_{v^*}^{w^*}$ are H -distinguishable, then $x_{j_1^*}, x_{j_2^*}, \dots, x_{j_r^*}$ are the least measurements that make the whole system (2.1) observable.

Remark 3.12. It is worth mentioning that the minimum measurement set of a PBN are not necessarily unique, we can choose the appropriate measurement set to make the system observable according to the actual demand and observation cost.

Remark 3.13. This paper takes into account all the feasible schemes for adding measurements required to make the system observable, and then obtains the least measurements needed by comparison. Our results provide more feasible schemes for observer design than previous ones.

4. An illustrative example

Example 4.1. Consider the apoptosis network modeled by PBNs in [25], which consists of the following four subnetworks:

$$\begin{cases} x_1(t+1) = f_1^\sigma(x_1(t), x_2(t), x_3(t)), \\ x_2(t+1) = f_2^\sigma(x_1(t), x_2(t), x_3(t)), \\ x_3(t+1) = f_3^\sigma(x_1(t), x_2(t), x_3(t)), \end{cases} \quad (4.1)$$

where $x_1 \in \mathcal{D}$ denotes the concentration level of apoptosis proteins, $x_2 \in \mathcal{D}$ denotes the state level presenting the active level of caspase 3, $x_3 \in \mathcal{D}$ presents the active level of caspase 8. The switching signal σ is a stochastic sequence taking values from $[1: 4]$, and

$$\begin{cases} f_1^1 = \neg x_2(t), \\ f_2^1 = \neg x_1(t) \wedge \neg x_3(t), \\ f_3^1 = x_2(t), \end{cases} \quad \begin{cases} f_1^2 = \neg x_2(t), \\ f_2^2 = x_2(t), \\ f_3^2 = x_2(t), \end{cases} \quad \begin{cases} f_1^3 = x_1(t), \\ f_2^3 = \neg x_1(t) \wedge \neg x_3(t), \\ f_3^3 = x_2(t), \end{cases} \quad \begin{cases} f_1^4 = x_1(t), \\ f_2^4 = x_2(t), \\ f_3^4 = x_2(t). \end{cases}$$

The probability distribution of each subnetwork is $\mathbb{P}\{f = f^1\} = 0.27$, $\mathbb{P}\{f = f^2\} = 0.03$, $\mathbb{P}\{f = f^3\} = 0.63$, $\mathbb{P}\{f = f^4\} = 0.07$, where $f^i = \{f_1^i, f_2^i, f_3^i\}$, $i \in [1: 4]$.

The output of system (4.1) is as follows:

$$y(t) = (x_2 \wedge \neg x_3) \vee (x_1 \leftrightarrow x_3 \wedge \neg x_2).$$

Identify $1 \sim \delta_2^1, 0 \sim \delta_2^2$. Then, the algebraic form of (4.1) can be expressed as

$$\begin{cases} x(t+1) = \mathbf{L}_{\sigma(t)}x(t), \\ y(t) = Hx(t), \end{cases} \quad (4.2)$$

where $x(t) = \bigotimes_{i=1}^3 x_i(t) \in \Delta_8$, $y(t) \in \Delta_2$, and $\mathbf{L}_i \in \mathcal{L}_{8 \times 8}$, $i \in [1: 4]$ is the structure matrix of the i -th subnetwork, $H \in \mathcal{L}_{2 \times 8}$, which are shown as

$$\begin{aligned} \mathbf{L}_1 &= \delta_8[7 \ 7 \ 4 \ 4 \ 7 \ 5 \ 4 \ 2], \quad \mathbf{L}_2 = \delta_8[5 \ 5 \ 4 \ 4 \ 5 \ 5 \ 4 \ 4], \\ \mathbf{L}_3 &= \delta_8[3 \ 3 \ 4 \ 4 \ 7 \ 5 \ 8 \ 6], \quad \mathbf{L}_4 = \delta_8[1 \ 1 \ 4 \ 4 \ 5 \ 5 \ 8 \ 8], \\ H &= \delta_2[2 \ 1 \ 1 \ 2 \ 2 \ 1 \ 2 \ 1]. \end{aligned} \quad (4.3)$$

According to the probability distribution of each subnetwork, we can get the probability distribution vector of (4.1) as $\mathbf{p}^\sigma = [0.27 \ 0.03 \ 0.63 \ 0.07]^\top$.

Let $\mathbf{L} = [\mathbf{L}_1 \ \mathbf{L}_2 \ \mathbf{L}_3 \ \mathbf{L}_4]$. The expectation of the augmented system can be calculated as

$$\mathbb{E}\{z(t+1)\} = \mathbf{Q}\mathbb{E}\{z(t)\},$$

where $\mathbf{Q} = \{[\mathbf{L}(I_{4 \times 2^3} \otimes \mathbf{I}_{2^3}^\top)] * [\mathbf{L}(I_4 \otimes \mathbf{I}_{2^3}^\top)]\} \mathbf{p}^\sigma$, and the columns of $\mathbf{Q}$ are shown in Table 2.

According to (2.9), it is easy to obtain that

$$\begin{aligned} S_0 &= \{\delta_{64}^1, \delta_{64}^{10}, \delta_{64}^{19}, \delta_{64}^{28}, \delta_{64}^{37}, \delta_{64}^{46}, \delta_{64}^{55}, \delta_{64}^{64}\}, \\ S_1 &= \{\delta_{64}^4, \delta_{64}^5, \delta_{64}^7, \delta_{64}^{11}, \delta_{64}^{14}, \delta_{64}^{16}, \delta_{64}^{22}, \delta_{64}^{24}, \delta_{64}^{29}, \delta_{64}^{31}, \delta_{64}^{48}\}, \\ S_2 &= \{\delta_{64}^2, \delta_{64}^3, \delta_{64}^6, \delta_{64}^8, \delta_{64}^{12}, \delta_{64}^{13}, \delta_{64}^{15}, \delta_{64}^{20}, \delta_{64}^{21}, \delta_{64}^{23}, \delta_{64}^{30}, \delta_{64}^{32}, \delta_{64}^{38}, \delta_{64}^{40}, \delta_{64}^{47}, \delta_{64}^{56}\}. \end{aligned}$$

The state evolutionary trajectories of S_1 can be shown in Fig. 1.

From formulas (3.5) and (3.7), one knows that $\mathcal{N}_1 = \{\delta_{64}^5, \delta_{64}^{14}, \delta_{64}^{24}, \delta_{64}^{31}\}$ and $\mathcal{P} = \{\delta_{64}^4, \delta_{64}^5, \delta_{64}^{29}\}$, then

$$\mathcal{N}_1 \cup \mathcal{P} = \{\delta_{64}^4, \delta_{64}^5, \delta_{64}^{14}, \delta_{64}^{24}, \delta_{64}^{29}, \delta_{64}^{31}\}.$$

It is known from Lemma 3.5 that to make PBN (4.1) observable, $\mathcal{N}_1 \cup \mathcal{P}$ must be H -distinguishable. Since δ_{64}^{13} and δ_{64}^{34} , δ_{64}^{15} and δ_{64}^{50} , δ_{64}^{29} and δ_{64}^{36} , δ_{64}^{31} and δ_{64}^{52} have the same distinguishability, respectively, we just consider states δ_{64}^{13} , δ_{64}^{15} , δ_{64}^{29} and δ_{64}^{31} . If $\mathcal{N}_1 \cup \mathcal{P}$

Table 2
Transition probability matrix of (4.2).

Column index j	j th column of $\mathbf{Q}$
$j \in S_1$	\
$j = 4$	$0.07\delta_{64}^4 + 0.63\delta_{64}^{20} + 0.03\delta_{64}^{36} + 0.27\delta_{64}^{52}$
$j = 5$	$0.07\delta_{64}^5 + 0.63\delta_{64}^{23} + 0.03\delta_{64}^{37} + 0.27\delta_{64}^{55}$
$j = 7$	$0.07\delta_{64}^8 + 0.63\delta_{64}^{24} + 0.03\delta_{64}^{36} + 0.27\delta_{64}^{52}$
$j = 11$	$0.07\delta_{64}^4 + 0.63\delta_{64}^{20} + 0.03\delta_{64}^{36} + 0.27\delta_{64}^{52}$
$j = 14$	$0.07\delta_{64}^5 + 0.63\delta_{64}^{21} + 0.03\delta_{64}^{37} + 0.27\delta_{64}^{53}$
$j = 16$	$0.07\delta_{64}^8 + 0.63\delta_{64}^{22} + 0.03\delta_{64}^{36} + 0.27\delta_{64}^{50}$
$j = 22$	δ_{64}^{29}
$j = 24$	$0.27\delta_{64}^{26} + 0.03\delta_{64}^{28} + 0.63\delta_{64}^{30} + 0.07\delta_{64}^{32}$
$j = 29$	$0.1\delta_{64}^{29} + 0.9\delta_{64}^{31}$
$j = 31$	$0.3\delta_{64}^{28} + 0.7\delta_{64}^{32}$
$j = 48$	$0.27\delta_{64}^{34} + 0.03\delta_{64}^{36} + 0.63\delta_{64}^{38} + 0.07\delta_{64}^{40}$

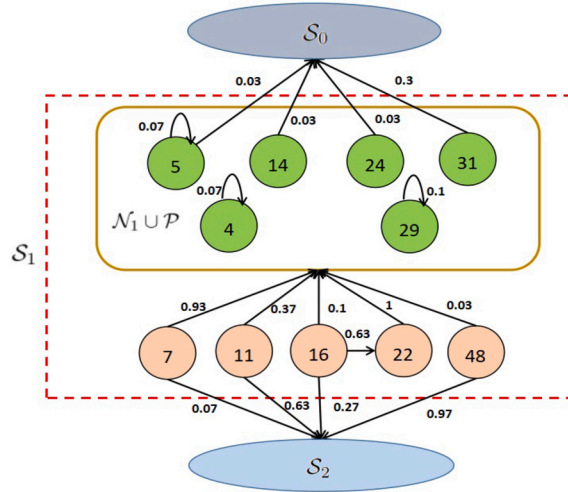


Fig. 1. The state transfer graph of S_1 .

becomes H -distinguishable, it can be seen that $\delta_{64}^7, \delta_{64}^{11}, \delta_{64}^{16}, \delta_{64}^{22}, \delta_{64}^{48}$ can reach the H -distinguishable states with probability one by Fig. 1, then all states in S_1 can be distinguishable. Thus, as long as all states in $\mathcal{N}_1 \cup \mathcal{P}$ become H -distinguishable, then PBN (4.1) is observable.

Next, we make $\mathcal{N}_1 \cup \mathcal{P}$ H -distinguishable by adding new measurements. We take the first state δ_{64}^4 as an example. Since δ_{64}^4 can be decomposed into $\delta_2^1 \delta_2^1 \delta_2^1$ and $\delta_2^1 \delta_2^2 \delta_2^2$, then we know that δ_{64}^4 becomes H -distinguishable under new measurements $y = x_2$ or $y = x_3$ based on formula (3.14) and (3.15), $\delta_{64}^4 \in Y_i, i \in \{2, 3\}$. So $\xi_1 = [0 \ 1 \ 1]^T$. Similarly, we can get that $\xi_2 = [1 \ 0 \ 0]^T, \xi_3 = [1 \ 0 \ 0]^T, \xi_4 = [1 \ 0 \ 1]^T, \xi_5 = [1 \ 1 \ 1]^T, \xi_6 = [1 \ 0 \ 0]^T$. It can be obtained that

$$\Phi = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

By Theorem 3.10, we can easily see that

$$\text{Row}_1(\Phi) \vee \text{Row}_i(\Phi) = \mathbf{1}_6^T, i \in \{2, 3\}.$$

Therefore, x_1, x_2 or x_1, x_3 are the least measurements that make PBN (4.2) observable. We can consider the cost of observing caspase 3 and caspase 8 to determine which measurements to be selected.

5. Conclusion

In this study, we have investigated the minimum observability of PBNs. Applying the STP method, we have established an augmented system and converted the observability problem of the original PBN into the set reachability problem of the augmented system. Then the minimum observability of the PBN has been solved by two main steps: determine the states that need to be H -distinguishable from the augmented system and add the least measurements to make a give state set H -distinguishable. For the first issue, we have found all the minimal H -distinguishable state sets which can make the originally unobservable PBN become observable. And for the second issue, we have proposed a necessary and sufficient condition for finding the least measurements to make a given set H -distinguishable. Furthermore, by comparing the numbers of measurements for all the feasible H -distinguishable state sets, we have obtained a method to determine all the least measurements to make the system observable. Compared with the existing minimum observable results, our method provide more feasible schemes for observer design.

We mainly apply the robust reachable set technique to find the least measurements. Although we greatly reduce the computational complexity by means of positive probability fixed points and invariant sets, it is still not a simple calculation process. In the future, we will consider the reducibility of measurements so that the computational complexity can be reduced further.

CRedit authorship contribution statement

Jiayi Xu: Writing – review & editing, Writing – original draft, Software, Data curation. **Shihua Fu:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Liyuan Xia:** Visualization, Investigation. **Jianjun Wang:** Visualization, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A

Proof of Lemma 3.5. We first prove that $\mathcal{N}_1$ must become H -distinguishable. Assume that there exists a state $z_0 \in \mathcal{N}_1$ which is not H -distinguishable. Since $z_0 \in \mathcal{N}_1$, there is a switching signal $\sigma^1 \in \Sigma^1$ such that $z(1; \sigma^1, z_0) \in S_0$. Obviously, for any integer $t \geq 1$, $z(t; \sigma^t, z_0) \in S_0$ holds. Suppose that $z_0 = x_0 \bowtie x'_0$, then $x(t; \sigma^t, x_0) \bowtie x'(t; \sigma^t, x'_0) \in S_0$ for any $t \in \mathbb{Z}_+$, which shows that $y(t; \sigma^t, x_0) = y'(t; \sigma^t, x'_0)$. So the PBN is unobservable. Therefore, $\mathcal{N}_1$ must be H -distinguishable.

Then we prove that $\mathcal{P}$ must be H -distinguishable. Suppose that there exists a positive-probability fixed point $z_0 \in \mathcal{P}$ which is H -indistinguishable. According to Definition 3.4, we can know that

$$\mathbb{P}\{z(t+1) = z_0 \mid z(t) = z_0\} > 0.$$

Hence, for any $t \in \mathbb{Z}_+$, z_0 can always reach itself with a positive probability. Since z_0 can not reach any other states with probability one, as long as z_0 is not H -distinguishable, it can not become a distinguishable state. So $\mathcal{N}$ can not become a distinguishable set. Thus, $\mathcal{P}$ must be H -distinguishable. The proof of the lemma is established. $\square$

Proof of Theorem 3.7. (Sufficiency) If the states in set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ are H -distinguishable, $v \in [1 : s], w \in [1 : c]$, then the states of $R(\Gamma)$ and $\mathcal{N}'$ are distinguishable. Since $\mathcal{N} = \mathcal{N}_1 \cup \mathcal{P} \cup (R(\Gamma) \cap \mathcal{N}') \cup \mathcal{N}'$, then all states in $\mathcal{N}$ are distinguishable. Therefore, PBN (2.1) is observable.

(Necessity) Suppose there is no set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ which is H -indistinguishable, $v \in [1 : s], w \in [1 : c]$, then $\mathcal{N}_1 \cup \mathcal{P}$ is not H -distinguishable, or $\mathcal{N}_1 \cup \mathcal{P}$ is H -distinguishable, but all sets of Θ_v^w are H -indistinguishable. If $\mathcal{N}_1 \cup \mathcal{P}$ is not H -distinguishable, then PBN (2.1) is unobservable by Lemma 3.5. If Θ_v^w is H -indistinguishable, then $\mathcal{N}'$ is indistinguishable. Thus PBN (2.1) is unobservable. In summary, to make PBN (2.1) become observable, at least one set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ must be H -distinguishable. $\square$

Proof of Theorem 3.10. (Necessity) Suppose that $x_{j_1}, x_{j_2}, \dots, x_{j_r}$ are the least measurements that make all states in $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ H -distinguishable, where $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w = \{z_1, z_2, \dots, z_l\}$. Then for any state $z_c \in \mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$, there exists at least one element $j_k \in \{j_1, j_2, \dots, j_r\}$ such that

$$(\Phi_v^w)_{j_k,c} = 1.$$

Since there are no fewer measurements that make all states in $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ H -distinguishable, therefore, $\{j_1, j_2, \dots, j_r\}$ is the set with the least number of elements that makes formula (3.16) hold.

(Sufficiency) Assume that $\mathcal{I}_v^w = \{j_1, j_2, \dots, j_r\}$ is the set with the least number of elements in $\{1, 2, \dots, n\}$ such that $\bigvee_{i=1}^r \text{Row}_{j_i}(\Phi_v^w) = \mathbf{1}_l^\top$. Then for any set $\{i_1, i_2, \dots, i_k\} \subseteq \{1, 2, \dots, n\}$, where $k < r$, it holds that $\bigvee_{a=1}^k \text{Row}_{i_a}(\Phi_v^w) \neq \mathbf{1}_l^\top$. This means that there is at least one state $z_\mu \in \mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ such that $(\Phi_v^w)_{i_v,\mu} = 0$ for any $i_v \in \{i_1, i_2, \dots, i_k\}$. Then state z_μ is indistinguishable under measurements $x_{i_1}, x_{i_2}, \dots, x_{i_k}$. This shows that $x_{i_1}, x_{i_2}, \dots, x_{i_k}$ cannot make all the states of set $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ become H -distinguishable. Therefore, $x_{j_1}, x_{j_2}, \dots, x_{j_r}$ are the least measurements that make all states in $\mathcal{N}_1 \cup \mathcal{P} \cup \Theta_v^w$ H -distinguishable. $\square$

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